

John Doe

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Education:

University of California, Berkeley | Berkeley, CA

GPA: 3.8/4.0

Bachelor of Engineering in Mechanical Engineering | May 2025

- Relevant Coursework: Thermodynamics, Fluid Mechanics, Heat Transfer, Solid Mechanics, Mechatronics, Control Systems, Manufacturing Processes, Finite Element Analysis

Skills:

- **CAD/CAE:** SolidWorks (CSWP), Autodesk Inventor, Fusion 360, ANSYS, Abaqus, COMSOL
- **Programming & Software:** Python (NumPy, Pandas), MATLAB, C++, LabVIEW, Simulink, EES
- **Manufacturing:** CNC Machining (G-code), 3D Printing (FDM, SLA), Laser Cutting, Welding, Mill, Lathe
- **Electronics:** Arduino, Raspberry Pi, Circuit Design, Soldering, Data Acquisition (DAQ)

Experience:

Mechanical Engineering Intern | Tesla | Palo Alto, CA | Summer 2024

- Designed and prototyped a new fixture for battery module testing, improving testing efficiency by 15%.
- Conducted thermal analysis on battery packs using ANSYS, identifying and mitigating potential overheating issues.
- Collaborated with a cross-functional team of engineers to troubleshoot and resolve manufacturing line discrepancies.

Undergraduate Researcher | UC Berkeley Robotics & Human Engineering Laboratory | Berkeley, CA | Fall 2023 - Spring 2024

- Contributed to the development of a lower-limb exoskeleton for gait rehabilitation.
- Utilized SolidWorks to model and iterate on mechanical components, optimizing for weight and strength.
- Programmed Arduino microcontrollers to control sensor and actuator systems.

Projects:

Formula SAE Racecar | Berkeley Formula Racing | Fall 2022 - Present

- Led the design and analysis of the suspension system for the 2024-2025 car, using SolidWorks and ANSYS for FEA.
- Manufactured and assembled suspension components using CNC machining and TIG welding.
- Integrated data acquisition systems to collect on-track performance data for design validation.
- Collaborated with a team of 30+ students in a fast-paced, competitive engineering environment.

Automated Cocktail Mixer | Personal Project | Spring 2023

- Designed and built a fully automated cocktail mixing machine using an Arduino Mega, peristaltic pumps, and a custom-built user interface.
- Modeled the enclosure and mechanical components in Fusion 360 and fabricated them using 3D printing and laser cutting.
- Wrote over 1,000 lines of C++ code to control the machine's functions, including ingredient selection, pouring, and mixing.
- The project was featured in the Berkeley Engineering annual project showcase.